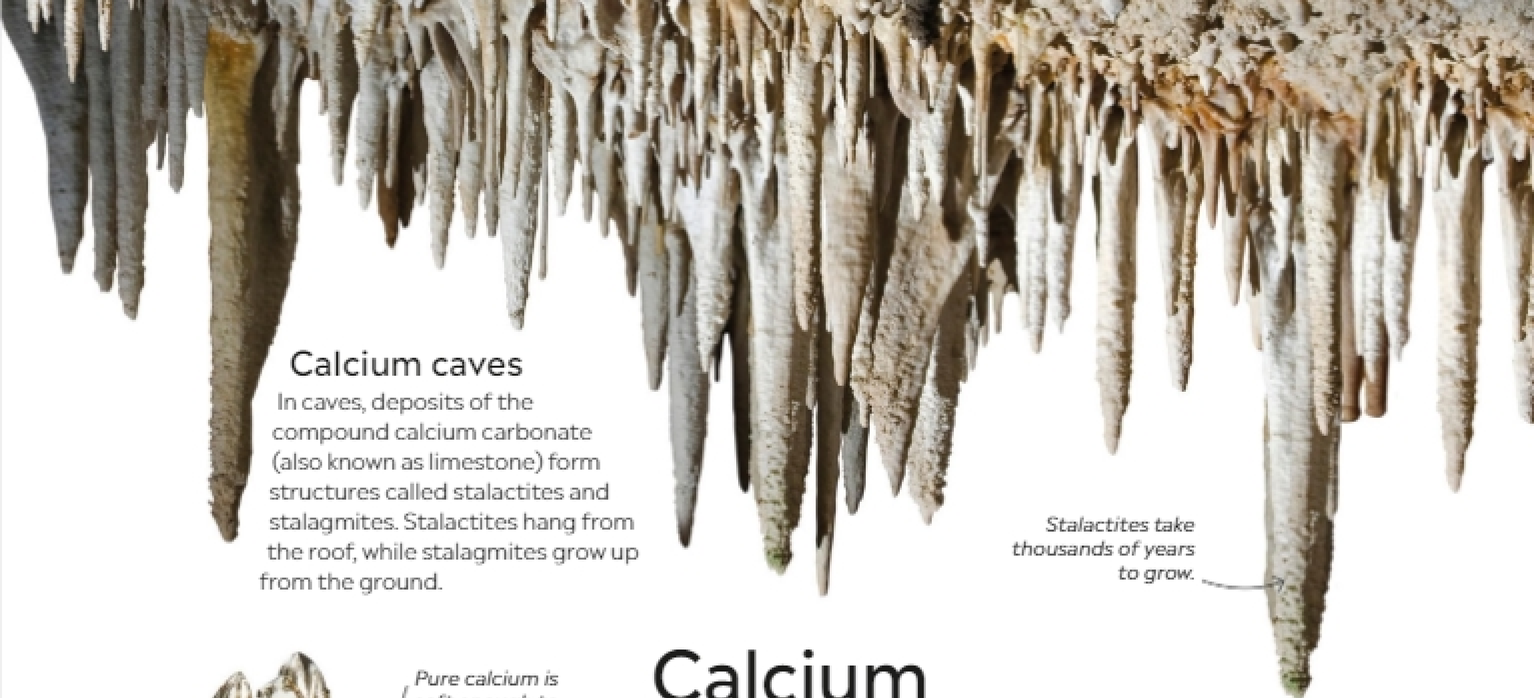


Calcium caves

In caves, deposits of the compound calcium carbonate (also known as limestone) form structures called stalactites and stalagmites. Stalactites hang from the roof, while stalagmites grow up from the ground.

Stalactites take thousands of years to grow.

Calcium

20 Ca

This soft metal is the fifth-most-abundant element in Earth's crust. As part of a mineral called hydroxylapatite, it strengthens our bones and teeth. Calcium compounds are also useful in industry and construction. For example, calcium oxide is used in the production of cement, while calcium hydroxide is used for treating sewage.

Pure calcium is soft enough to cut with a knife.

Crystals of pure calcium refined in a laboratory

Acid controller

Calcium compounds are used to reduce acidity in various ways. They are added to tablets that reduce acid-related discomfort in the stomach. Calcium supplements can also boost the growth of some plants by making the soil less acidic.

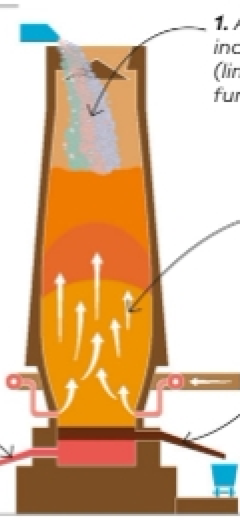
Indigestion tablets

Teeth and bones

Calcium is the most abundant metal in our body. Calcium isn't only needed for strong teeth and bones—it also helps our cells work properly. We get calcium in our diet by eating calcium-rich food, including dairy products and nuts.

PRODUCING IRON

Calcium carbonate (limestone) is important in the production of pure iron. It is added to the raw materials in a blast furnace, where it removes sand and other impurities from the iron ore.



1. *A mixture of raw materials, including calcium carbonate (limestone), is added to the furnace through this valve.*

2. *Hot air enters the furnace, heating up the materials inside.*

3. *Limestone mixes with the impurities in the ore to form a mixture called slag. This slag floats above the molten iron and is collected from the furnace.*

4. *Pure molten iron sinks to the bottom and is then drained from the furnace.*